

Electronic Supplementary Material: Evaluating design-stage manufacturability decisions against the resolution of production

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Abstract

This supplement gives the computation details of the paper and the further results it refers to: the content and protocol of the production floor, the measurement and the runs (Section S2), the distances of all rules with their intervals, one-sided values, paired differences and coverage (Section S3), the learners (Section S4), the procedure, the guards and the requirement scenarios (Section S5) and the robustness analysis (Section S6). Every table is written by the released code from its result files; section, table and figure numbers without the prefix S refer to the paper.

Keywords: design for manufacturing, operating characteristic, production floor, deep drawing

S1 Computation details

Feature extraction. Scans are reduced to a grid of 2×4-pixel medians after a luminescence filter (connected components of at least 20,040 pixels); calibration 0.0769 mm per pixel across the scan line, 0.1581 mm per line and 0.00775 mm per height unit. The cup-bottom plane is fitted robustly to the plateau at least 10 mm inside its edge, and all depths are taken from it. The flange comprises the points 20–40 mm below the bottom plane, at least 8 mm from the cup wall and 3 mm from the outer edge. Wall positions are interpolated where the binned (0.5 mm) radial profile reaches 10, 15 and 20 mm depth; the wall angle is taken between 10 and 20 mm. The arm angle is fitted to

the longest run of the profile whose slope stays below 15° , shortened by 1 mm at both ends and at least 4 mm long. Flange widths are medians over bands ± 2 mm about the mid-lines; corner tips are the outline points within 0.3 mm of the extreme diagonal projection. Simulated parts are mirrored from the quarter model and processed with the same definitions on a 1 mm grid.

Floors and effects. Batches of $B = 50$ consecutive parts; batches with fewer than 80 % valid values are dropped; centres are 10 % trimmed means; $\alpha = 0.95$. Intervals: 2000 resamples of the batches within alternatives (seed 1), pairs of copies of one batch excluded, and 2000 resamples of whole alternatives. Scatter floor: 50 random permutations of the parts within each series. Components: rational subgroups of five consecutive parts; successive-difference standard deviation $\sqrt{\Delta^2/2}$. Protocol (Table S3): σ_w is the pooled standard deviation within batches of 50 and σ_b the standard deviation of the batch means corrected for $\sigma_w^2/50$. Lag floors (Table S4): the 95 % quantile of the differences between batch centres a given number of batches apart. Between-series floor: the 95 % quantile of the differences between batch centres of different series of the same geometry and blank-holder force.

Decision rules. Adequacy share $p = 0.95$; per-geometry floors; requirements below zero are left out for these non-negative characteristics. The largest-residual margin is the $\lceil (n + 1) \cdot 0.9 \rceil$ -th smallest absolute residual of n calibration alternatives, the largest when this exceeds n . M1s: ordinary least squares of q_{95} on the nominal simulation s_0 over the calibration alternatives, M1 when fewer than two alternatives or no spread of s_0 . M5 and M5n: Gaussian-process regression with a constant times anisotropic squared-exponential kernel plus white noise, normalised targets, three optimiser restarts (random state 0), length scales in $[0.1, 100]$; the noise variance is bounded below by the larger of the mean squared block-bootstrap standard error of the calibration alternatives' q_{95} and the pooled variance of the targets between the calibration alternatives of one geometry and blank-holder force, relative to the variance of the targets; descriptors are a convex-geometry indicator, the blank-holder force in units of 100 kN and the lubrication rank (coarse 0, medium 1, fine 2, the order of the oil amount), with descriptors that are constant over the calibration set dropped. NN: mean q_{95} of the calibration alternatives of the target's geometry at the nearest blank-holder force (both neighbours when equally near; the other geometry when the family has none). Mc: for each lubrication pattern, the friction coefficient of the DDACS grid at 0.99 mm whose simulated mid-side and corner draw-in, in floors, are closest in least squares to the centres of the calibration alternatives with that pattern (all calibration alternatives if none has it). M3 and M3n (M4 in the code): histogram gradient boosting (maximum depth 3, 200 iterations, learning rate 0.08, random state 0) on the geometry indicator, blank-holder force, two lubrication indicators, sheet thickness and oil film; 500 incoming conditions are drawn for the new alternative from calibration parts with the same lubrication pattern, with in-sample residuals pooled over the calibration alternatives resampled. Their jackknife+ interval (Barber et al, 2021) uses the n leave-one-out fits over the calibration alternatives: each predicts the new alternative and leaves a residual on the alternative it left out; with the median residual as offset and the absolute deviations of the residuals from it as scores, the interval runs

from the $\lfloor 0.1(n+1) \rfloor$ -th smallest prediction minus score to the $\lceil 0.9(n+1) \rceil$ -th smallest prediction plus score, the extreme values when n is too small.

Distances and intervals. The shares of correct and wrong verdicts are evaluated exactly at the exclusion distances of the cases, on a grid of 0.05 floors up to 600 floors. Intervals: 1000 resamples of the held-out alternatives within each geometry (seeds derived from the scope), each with one of 1000 block-bootstrap replicates of every alternative's q_{95} (batches of 50 resampled within the series); the resamples are shared by all rules of a scope. Refitting bootstrap: 200 resamples in which the three lubrication patterns of each geometry are drawn with replacement (at least two distinct ones), all forces kept; every alternative of a drawn pattern enters as often as its pattern was drawn, a target's copies never calibrate it, and every alternative-level rule is refitted, so that each case keeps its evidence relation. The decomposition of Section 5.4 is a two-way analysis of variance of the log decisive distance over the eleven calibrated rules and the relations, without replication, so the interaction is the residual.

Procedure. The guard band g of a rule and relation is the smallest margin on a grid of 0.25 floors up to 20 floors beyond which at least 95 % of the decided verdicts are correct, the verdicts being those at the requirement grid of the paper (at the truth and at 0.5 to 10 floors in steps of 0.5, then 12, 15 and 20 floors, on both sides of it); the margin of a verdict is $(R - U)/F$ for meets and $(L - R)/F$ for fails. Step 6 widens the rule's interval by g floors on both sides, and its distances are computed with the exact estimator. The guard bands are chosen and evaluated on the same cases.

Budget, guard, scenarios, robustness. The budget uses all subsets of $k = 1, \dots, 8$ calibration alternatives of the family (M5 and M5n from $k = 2$), bands from 200 resamples of the held-out alternatives. The guards and the tolerance scenarios use the stored intervals of every rule. ISO 2768-1 angular rows by the shorter leg: the wall (about 28 mm) and the concave arm (about 18 mm) over 10 up to 50 mm, the convex arm (about 9 mm) up to 10 mm. Capability-anchored requirements: the upper limit at the centre of the parts plus $3P_{pk}$ long-term standard deviations. The temperature correction subtracts $b(T - \bar{T}_a)$ from each characteristic, with b the pooled within-alternative slope on the punch temperature T . The smoothed simulation replaces the simulated values of each geometry and force by a quadratic response surface in sheet thickness and friction coefficient. Calibration series of n parts: the calibration alternatives' q_{95} and its standard error from their first n parts, the truth from all 500. First 150 parts dropped: floors, truths and calibrations from parts 151–500 of every series, after the warm-up of the punch. Floor between series: every distance rescaled with the floor taken between the series of one geometry and force. Unit of q_{95} reproducibility: every distance rescaled with the 95 % quantile of the difference between the q_{95} of two batches of 100 parts.

Software. Python 3.11 with NumPy 2.4, pandas 3.0, SciPy 1.17, scikit-learn 1.9 (Pedregosa et al, 2011) and Matplotlib 3.11.

S2 Production floor, measurement and runs

Table S1 defines the quality characteristics. Table S2 decomposes the floor and gives the reproducibility of the decided statistic, Table S3 the variance components that

Table S1 Quality characteristics, extracted with the same definitions from the scans and the simulated nodes

Characteristic	Unit	Role	Definition
Draw-in, mid-side	mm	process indicator	$(210 - W_x)/2$; W_x : flange width across the cup centre, across the scan lines
Draw-in, corner	mm	process indicator	$(2\sqrt{2} \cdot 210 - L_1 - L_2)/4$; L : distance between opposite corner tips of the flange
Flange waviness	mm	process indicator	robust standard deviation of the flange about a fitted quadratic surface
Wall angle	°	drawing	inclination of the wall from the punch axis between 10 and 20 mm depth, mean of the east and west walls (design angle 20° concave, 10° convex)
Arm angle, cut	°	drawing	inclination of the flat segment of the cut arms, mean of four arms (requirement on its absolute value)
Cup depth	mm	drawing	median depth of the flange below the plane of the cup bottom after drawing
Bottom dome, cut	mm	drawing	sag of the bottom at 40 mm radius from a fitted paraboloid (requirement on its absolute value)

All depths are referred to the plane of the cup bottom; definitions of the extraction in Section S1.

Table S2 What the production floor contains: components and variants in floors

Characteristic	σ_{LT}/σ_{ST}	White share (%)	First half	q_{95} floor	Between series	
					median	95 %
Draw-in, mid-side	1.22	24	0.93	1.49	0.9	2.4
Draw-in, corner	1.24	17	0.63	1.08	0.7	2.4
Flange waviness	1.16	23	0.81	1.07	0.5	1.2
Wall angle	1.23	33	0.90	0.92	0.5	2.7
Arm angle, cut	1.35	23	0.71	0.96	1.0	4.3
Cup depth	1.41	15	0.95	1.11	0.5	4.8
Bottom dome, cut	1.10	17	0.49	1.02	1.6	9.8

σ_{ST} : standard deviation within subgroups of five consecutive parts; σ_{LT} : of the whole series (medians). White share: the floor that independent part-to-part noise alone would produce, $1.96\sqrt{2/B}$ times the successive-difference standard deviation, over F ; it bounds the share a scanner's repeatability can explain. First half: floor of the first 250 parts. q_{95} floor: 95 % quantile of the difference between the 95th percentiles of two batches of 100 parts. Between series: difference between the centres of the three lubrication series of one geometry and force.

allow the floor to be recomputed for another protocol, and Table S4 its dependence on the time between the batches compared. Table S5 counts the single-factor contrasts that production resolves, and Fig. S1 shows the alternatives against the floor. Table S6 gives the offsets of the nominal simulation and Table S7 the extraction noise of the simulated characteristics and the accuracy of a surrogate of the simulations. Tables S8 and S9 document the measurement and Table S10 the run conditions.

Table S3 Floor protocol: variance components and the floor in units of the part scatter

Characteristic	Geometry	F	σ_{LT}	σ_{ST}	σ_w	σ_b	F/σ_{LT}	F/σ_{ST}	model F
Draw-in, mid-side	concave	0.050	0.041	0.038	0.045	0.016	1.21	1.30	0.048
	convex	0.052	0.034	0.027	0.034	0.018	1.53	1.94	0.051
Draw-in, corner	concave	0.053	0.031	0.023	0.034	0.016	1.69	2.27	0.047
	convex	0.030	0.023	0.020	0.023	0.010	1.31	1.47	0.030
Flange waviness	concave	0.0037	0.0017	0.0012	0.0020	0.0012	2.15	3.07	0.0034
	convex	0.0021	0.0027	0.0027	0.0028	0.00064	0.79	0.80	0.0021
Wall angle	concave	0.081	0.050	0.046	0.055	0.025	1.60	1.75	0.074
	convex	0.078	0.12	0.11	0.11	0.020	0.66	0.68	0.070
Arm angle, cut	concave	0.12	0.061	0.048	0.054	0.042	1.89	2.44	0.12
	convex	0.100	0.11	0.093	0.12	0.031	0.93	1.07	0.098
Cup depth	concave	0.0087	0.0039	0.0029	0.0057	0.0027	2.22	3.02	0.0077
	convex	0.012	0.0074	0.0051	0.0064	0.0041	1.67	2.42	0.012
Bottom dome, cut	concave	0.0034	0.0024	0.0019	0.0026	0.0011	1.44	1.86	0.0033
	convex	0.0082	0.0054	0.0042	0.0045	0.0030	1.54	1.97	0.0084

Units of the characteristic. σ_w : pooled standard deviation within batches of 50; σ_b : between batch means, corrected for $\sigma_w^2/50$; model F : $1.96\sqrt{2}\sqrt{\sigma_b^2 + \sigma_w^2/50}$. Medians over the nine series.

Table S4 The floor by the time between the batches compared, in floors

Lag (parts)	DM	DC	Wav	Wall	Arm	Depth	Dome
1 (50)	0.80	0.60	0.51	0.65	0.65	0.52	0.55
2 (100)	0.82	0.77	0.79	0.87	0.89	0.79	0.69
3 (150)	0.99	0.85	0.92	0.90	0.95	0.85	0.81
4 (200)	1.04	0.97	1.00	1.00	1.06	0.86	0.94
5 (250)	0.98	1.17	1.21	1.02	1.13	1.02	1.08
6 (300)	1.04	1.29	1.26	1.22	1.27	1.10	1.24
7 (350)	0.98	1.30	1.41	1.23	1.37	1.13	1.30
8 (400)	0.98	1.13	1.63	1.30	1.51	1.28	1.45
9 (450)	1.22	1.10	1.54	1.42	1.43	1.39	1.50

95% quantile of the difference between batch centres a given number of batches apart, over the floor (mean of the two geometries). DM, DC: mid-side and corner draw-in; Wav: flange waviness.

Table S5 Single-factor contrasts that production resolves and minimum resolvable force change

Characteristic	Geometry		Blank-holder force		Lubrication		Force MRC (kN)	
	ESR ≥ 1	Median	ESR ≥ 1	Median	ESR ≥ 1	Median	100–300	300–500
Draw-in, mid-side	9/9	18.7	18/18	6.8	9/18	0.9	32	105
Draw-in, corner	9/9	108.8	18/18	4.3	7/18	0.7	67	47
Flange waviness	9/9	2.8	15/18	6.4	3/18	0.5	21	78
Wall angle	9/9	129.0	4/18	0.6	5/18	0.5	>200	>200
Arm angle, cut	9/9	47.2	17/18	3.0	9/18	1.0	48	>200
Cup depth	9/9	34.6	18/18	10.2	7/18	0.5	21	43
Bottom dome, cut	9/9	9.6	15/18	2.4	11/18	1.6	>200	>200
All	63/63	34.6	105/126	4.3	51/126	0.7		

ESR: effect-to-scatter ratio of two alternatives that differ in one factor. MRC: floor over the local sensitivity of the alternative centres to blank-holder force, between 100 and 300 kN (the stroke speed also changes) and between 300 and 500 kN; >200: beyond the interval.

Table S6 Offset between the parts' 95th percentile and the nominal simulation, in floors

Characteristic	Concave		Convex		Matched, signed	
	Offset	Spread	Offset	Spread	Concave	Convex
Draw-in, mid-side (mm)	−16	37	−20	32	−0.60	−1.19
Draw-in, corner (mm)	−59	17	−108	22	−2.96	−3.32
Flange waviness (mm)	7	11	2	13	+0.03	−0.01
Wall angle (°)	16	19	12	13	+1.20	+1.03
Arm angle, cut (°)	−27	39	1	15	+4.26	−0.04
Cup depth (mm)	−109	14	−74	31	−0.95	−0.90
Bottom dome, cut (mm)	−28	22	10	9	−0.04	+0.06

Offset: median over the nine alternatives of a geometry of $(q_{95} - s_0)/F$, s_0 the nominal simulation, on the decision scale; spread: its range. Matched, signed: median difference between the centre of the parts and of their matched simulations, in the unit, with signs. Removing the cup-depth reference offset brings the decisive distance of M0 from 108 to 65 floors.

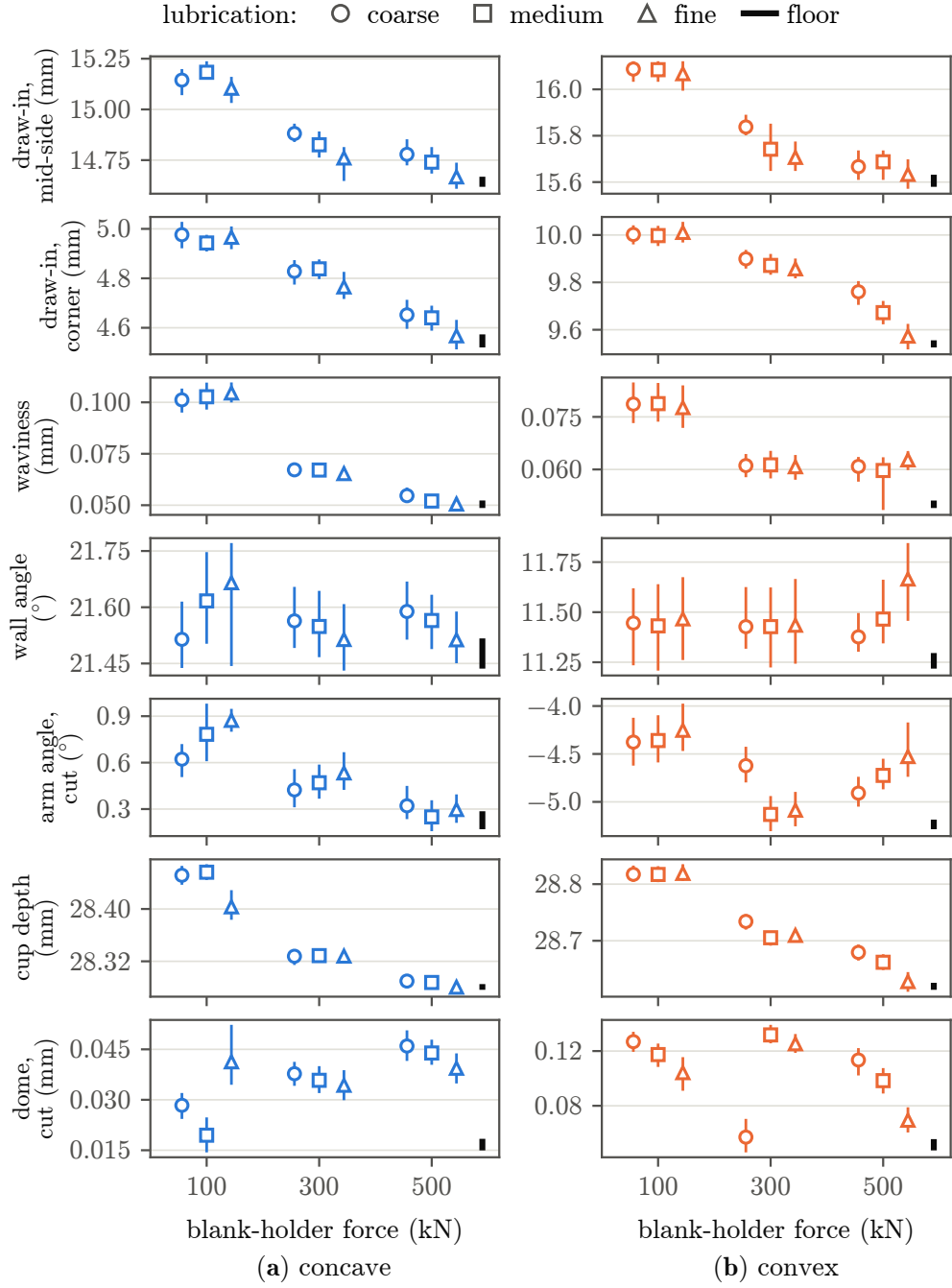


Fig. S1 Centre and 5–95 % range of each characteristic per alternative; the black bar is the production floor

Table S7 Extraction noise of the simulated characteristics and accuracy of a surrogate (floors)

Characteristic	Concave			Convex		
	noise	envelope	surrogate	noise	envelope	surrogate
Draw-in, mid-side	0.1	0.8	0.1	0.1	0.7	0.1
Draw-in, corner	0.1	0.5	0.0	0.2	1.3	0.1
Flange waviness	0.4	2.9	0.8	4.0	4.2	5.6
Wall angle	3.1	2.0	0.7	1.7	1.5	1.8
Arm angle, cut	0.9	1.5	0.9	0.9	2.5	1.8
Cup depth	0.8	1.5	0.8	0.7	1.6	1.1
Bottom dome, cut	9.8	20.9	11.2	2.4	9.9	1.7

Noise: standard deviation of the second difference along the friction grid over $\sqrt{1.5}$. Envelope: largest difference between the maximum of the simulated values of a geometry and force and that of a quadratic response surface. Surrogate: leave-one-out error of a Gaussian-process surrogate of the simulations over process conditions.

Table S8 Redundancy of the scans: ranges over the nine series of each geometry

Quantity	Concave	Convex
Flange width W_x , SD (mm)	0.07–0.15	0.05–0.11
Flange width W_y , SD (mm)	0.27–0.54	0.43–0.92
Correlation of W_x and W_y	−0.14 to 0.20	−0.21 to 0.09
$W_y - W_x$ (mm)	2.5–3.5	2.3–3.5
Wall angle, mean of E and W, SD (°)	0.04–0.10	0.06–0.13
Wall angle N, SD (°)	0.31–0.35	0.35–0.38
Wall angle N minus mean of E and W (°)	0.0–0.4	−1.6 to −0.9
Correlation of the E and W wall angles	−0.10 to 0.27	−0.71 to −0.49

W_x, W_y : flange width across the cup centre, across and along the scan lines; E, W, N: east, west and north walls; SD and correlation over the parts of a series.

Table S9 Measurement resolution of the characteristics

Characteristic	Step (unit)	Step/ F	Distinct values per series	Zero SE of q_{95}
Draw-in, mid-side	0.019	0.378	10 (6–21)	5
Draw-in, corner	0.00014	0.003	497 (491–500)	0
Flange waviness	0.0000100	0.003	480 (469–494)	0
Wall angle	0.00039	0.005	499 (497–500)	0
Arm angle, cut	0.00049	0.004	500 (496–500)	0
Cup depth	0.000035	0.003	493 (486–497)	0
Bottom dome, cut	0.000021	0.004	490 (474–497)	0

Step: median difference between adjacent distinct values within a series. Zero SE: alternatives whose block-bootstrap standard error of q_{95} is zero (the quantile does not move).

Table S10 Run metadata of the 18 series

Geometry	Force (kN)	Oiling	Punch temperature			Sheet (mm)	Speed (mm/s)
			start (°C)	rise, 150 (K)	rise, all (K)		
concave	100	coarse	20.5	4.6	6.7	0.991	47
concave	100	fine	23.0	4.6	6.4	0.987	47
concave	100	medium	23.0	3.4	3.7	0.993	47
concave	300	coarse	23.6	4.3	5.7	0.992	79
concave	300	fine	27.2	2.6	1.9	0.992	79
concave	300	medium	25.7	3.5	3.6	0.993	79
concave	500	coarse	22.6	5.0	6.1	0.992	79
concave	500	fine	21.9	3.3	6.8	0.992	79
concave	500	medium	25.6	3.3	4.8	0.992	79
convex	100	coarse	26.5	1.5	2.7	0.986	47
convex	100	fine	28.1	1.1	1.5	0.987	47
convex	100	medium	28.2	1.0	1.1	0.986	47
convex	300	coarse	23.9	4.0	5.9	0.990	78
convex	300	fine	26.9	1.7	2.6	0.986	78
convex	300	medium	21.9	4.1	6.6	0.986	78
convex	500	coarse	24.7	3.7	5.7	0.991	78
convex	500	fine	29.6	1.5	2.0	0.990	78
convex	500	medium	27.5	2.5	3.3	0.990	78

Punch temperature at the start (median of the first ten parts) and its rise over the first 150 parts and over the series; median sheet thickness; median forming speed. The order and dates of the series are not documented.

S3 Distances

Table S11 gives the weight of the failing side in the estimator of Section 3.4. Table S12 gives the distances of all rules and scopes, Table S13 their bootstrap intervals, Table S14 the one-sided safe distances and Table S15 the paired differences between rules, with the fits held fixed and refitted. Table S16 gives the coverage and width of the interval rules and the accuracy of their centres, Tables S17 and S18 the distances per characteristic, in floors and in units, Table S19 the distances of a new family in the floor of the produced family, with a baseline from the drawing nominal, Table S20 the calibration budget behind Fig. 3 and Table S21 the subsets and variants of the cases discussed in Section 5.4.

Table S11 Weight of the failing side at given distances

δ (floors)	Feasible failing requirements	Weight of the failing side
1	126 of 126	0.50
3.4	124 of 126	0.50
8.5	115 of 126	0.48
16	98 of 126	0.44
35	84 of 126	0.40
108	63 of 126	0.33

A requirement $q_{95} - \delta F$ below zero is no requirement for a non-negative characteristic; every feasible verdict has equal weight (Section 3.4 of the paper).

Table S12 Decisive and safe distances (floors) of all rules and scopes

Rule	New	New process setting			New	Pooled	
	variant	all	interpolated	extrapolated	family	variant	setting
M0	108	108	109	106	108	108	108
M0w	119/94	119/94	119/94	119/77	119/94	119/94	119/94
M1	15	24	11	25	35	27	28
M1s	4.9	19	19	19	122	15	20
Mc	27	39	14	40	38	32	33
M2	23/0.15	18/8.1	18/2.7	18/9.9	51/16	53/0	53/1.0
M3	23/0	31/16	29/13	33/17	49/21	25/0	32/5.1
M5	7.2/0.75	19/6.8	19/2.8	19/8.9	38/28	11/0.05	28/5.4
M1n	9.1	10	4.8	12	131	130	130
NN	3.4	8.5	4.7	9.5	131	3.4	8.5
M2n	19/0	15/5.9	13/0	16/9.1	132/129	325/0	325/4.0
M3n	11/0.05	13/7.4	15/7.0	13/7.8	133/128	15/0	15/4.9
M5n	7.2/0.55	17/6.1	16/0	18/8.6	132/130	9.8/0.20	16/2.7
M6	17/0.05	56/6.2	47/0	56/8.1	41/25	–	–

δ_d/δ_s ; one number for rules that always decide. Pooled: the other geometry’s alternatives added to the calibration set. M3n is M4 in the code. M6: M5 mean with a conformal margin on standardised leave-one-out residuals.

Table S13 Bootstrap 95 % intervals of the decisive and safe distances (floors)

Rule	New variant		New setting		New family	
	δ_d	δ_s	δ_d	δ_s	δ_d	δ_s
M0	101–115	101–115	101–115	101–115	101–115	101–115
M0w	118–119	77–100	118–119	77–100	118–119	77–100
M1	12–18	12–18	21–25	21–25	33–37	33–37
M1s	4.7–6.3	4.7–6.3	14–21	14–21	121–123	121–123
Mc	24–29	24–29	34–40	34–40	36–45	36–45
M2	22–23	0–0.65	17–18	6.4–9.9	50–52	14–23
M3	17–26	0–0.05	25–36	13–17	45–53	16–28
M5	6.6–8.4	0–1.9	18–20	4.8–8.9	35–39	25–31
M1n	5.5–10	5.5–10	9.0–12	9.0–12	129–131	129–131
NN	2.0–4.5	2.0–4.5	5.5–10	5.5–10	129–131	129–131
M2n	15–21	0–0.70	15–16	3.3–9.3	131–133	127–129
M3n	10–13	0–0.90	13–15	5.5–9.3	131–133	126–129
M5n	6.4–7.5	0–1.7	16–18	2.5–9.0	130–133	128–130

1000 resamples of the held-out alternatives within each geometry, each with a block-bootstrap replicate of the true q_{95} ; the fits are held fixed (refitting: Table S29).

Table S14 One-sided safe distances (floors): false accepts and false rejects

Rule	New variant		New setting		New family	
	FA side	FR side	FA side	FR side	FA side	FR side
M0	17	109	17	109	17	109
M0w	9.2	100	9.2	100	9.2	100
M1	15	12	24	23	33	36
M1s	4.7	4.9	164	12	122	122
Mc	28	23	39	34	42	37
M2	0.30	0	8.3	6.7	11	23
M3	0	0	13	17	20	21
M5	0.15	0.85	7.6	5.5	32	27
M1n	10	5.2	12	9.2	173	129
NN	2.8	3.4	10	5.3	169	128
M2n	0.20	0	9.2	3.3	165	127
M3n	0	0.05	9.2	6.9	164	126
M5n	0.10	0.90	9.0	2.3	167	127

FA side: smallest distance beyond which at most 5 % of the verdicts on failing alternatives are false accepts; FR side: the same for false rejects of adequate alternatives.

Table S15 Paired differences of the decisive distance between rules (floors)

Pair	New variant	New setting	New family
M5 – M5n	0.0 (–0.4, +1.1)	+1.5 (–0.1, +3.0)	–93.8 (–95.8, –92.0)
M2 – M2n	+3.9 (+0.4, +7.4)	+2.3 (+1.4, +3.2)	–81.7 (–82.5, –78.5)
M1 – M1n	+6.2 (+4.6, +9.7)	+13.5 (+10.6, +14.9)	–95.4 (–96.6, –92.5)
M1s – M1n	–4.2 (–5.4, –0.4)	+8.9 (+3.2, +11.8)	–8.4 (–8.9, –6.7)
M1s – M1	–10.4 (–12.8, –7.0)	–4.6 (–10.2, –0.6)	+87.0 (+85.2, +89.3)
M1s – NN	+1.5 (+0.4, +4.0)	+10.7 (+5.0, +15.3)	–8.2 (–9.0, –7.2)
M5 – NN	+3.8 (+2.8, +5.2)	+10.2 (+8.2, +13.9)	–92.1 (–94.6, –91.0)
M5n – NN	+3.8 (+2.6, +5.1)	+8.8 (+6.2, +12.8)	+1.7 (+0.7, +1.7)
M2 – NN	+19.3 (+17.2, +20.7)	+9.1 (+6.9, +12.5)	–79.9 (–81.1, –77.5)
M5 – M2	–15.5 (–16.1, –13.0)	+1.2 (+0.5, +2.1)	–12.3 (–14.8, –11.3)
M5 – M3n	–4.2 (–6.4, –2.8)	+5.4 (+3.8, +6.6)	–94.7 (–97.2, –93.2)
M3 – M3n	+11.8 (+4.4, +15.7)	+17.4 (+11.6, +22.5)	–84.4 (–87.6, –79.8)
Mc – M1	+11.5 (+8.9, +12.9)	+14.8 (+13.4, +16.3)	+2.4 (+0.9, +9.1)
<i>Refitting bootstrap (lubrication patterns resampled)</i>			
NN – M5	–3.8 (–7.5, –1.8)	–10.2 (–12.1, –9.1)	+92.1 (+91.3, +95.9)
NN – M5n	–3.8 (–5.0, –2.0)	–8.8 (–10.3, –8.0)	–1.7 (–2.1, –0.5)
M5 – M5n	0.0 (–0.5, +3.5)	+1.5 (–0.2, +2.7)	–93.8 (–96.4, –92.6)
M2 – M2n	+3.9 (+2.7, +4.6)	+2.3 (+0.8, +2.3)	–81.7 (–82.6, –79.0)
M1s – NN	+1.5 (+1.5, +5.0)	+10.7 (+9.6, +12.4)	–8.2 (–14.1, –0.1)
M1 – M1n	+6.2 (+5.6, +7.7)	+13.5 (+12.4, +14.3)	–95.4 (–97.1, –93.6)

Difference and 95 % interval; a negative difference means that the first rule decides closer to the requirement. Upper block: common resamples of Table S13 (fits fixed); lower block: refitted rules.

Table S16 Interval rules: coverage of the true q_{95} , half-width and the accuracy of the centre

Rule	New variant			New setting			New family			Pooled variant		
	cov.	half	centre	cov.	half	centre	cov.	half	centre	cov.	half	centre
M0w	37	12	105	37	12	105	37	12	105	37	12	105
M2	89	7.6	8.6	53	5.1	12	47	7.5	34	94	11	16
M3	96	9.3	9.9	43	8.5	21	40	7.8	35	96	11	8.8
M5	86	2.2	3.3	55	5.2	12	12	1.9	34	90	2.5	2.9
M2n	90	5.9	9.1	53	4.1	10	12	5.5	131	91	32	130
M3n	89	3.0	3.7	44	2.6	10	7	3.1	131	94	4.0	5.0
M5n	85	2.3	3.1	55	4.4	11	0	2.0	131	89	2.3	3.0

cov.: share of cases whose true q_{95} lies in the interval (%; nominal 90 %; the largest residual attains at most $n/(n+1)$); half: median half-width (floors); centre: decisive distance of the interval centre used as a point rule (floors).

Table S17 Decisive/safe distance per characteristic (floors)

Characteristic	New variant					New setting				
	M1s	M2	M5	M5n	NN	M1s	M2	M5	M5n	NN
Draw-in, mid-side	2.2	20/0	3.4/0.15	3.7/0.10	2.0	4.0	18/8.3	20/6.8	10/5.1	6.5
Draw-in, corner	4.3	6.1/0.50	5.7/1.5	6.1/1.1	4.5	2.9	6.6/2.2	5.7/2.0	15/5.2	6.7
Flange waviness	7.2	23/0.05	1.9/0.20	1.8/0.20	0.85	21	16/11	19/11	18/9.7	11
Wall angle	3.0	9.3/0.15	4.8/1.8	3.6/0.65	3.4	2.0	7.6/3.2	7.6/3.2	3.1/1.3	1.9
Arm angle, cut	4.2	27/0.30	7.2/1.2	7.2/1.3	2.8	30	27/8.1	34/8.1	9.2/2.5	5.9
Cup depth	9.0	28/0.30	5.4/0.75	5.4/0.65	3.3	165	15/13	17/13	23/13	15
Bottom dome, cut	5.0	15/0.60	12/5.3	9.8/3.1	7.0	14	18/2.8	17/3.5	9.3/1.5	4.5

18 held-out alternatives per characteristic and scope; one wrong verdict moves a share by about three percentage points.

Table S18 Decisive distances per characteristic in its unit

Characteristic	New variant		New setting		New family		
	NN	M5	NN	M5	M1	M2	M5
Draw-in, mid-side (mm)	0.10	0.20	0.35	1.0	1.2	1.0	0.30
Draw-in, corner (mm)	0.15	0.20	0.25	0.25	0.75	0.35	0.35
Flange waviness (mm)	0.050	0.050	0.050	0.10	0.10	0.15	0.050
Wall angle (°)	0.30	0.40	0.15	0.65	1.4	0.80	0.60
Arm angle, cut (°)	0.30	0.75	0.60	4.0	4.1	5.3	4.0
Cup depth (mm)	0.050	0.10	0.15	0.20	0.30	0.30	0.20
Bottom dome, cut (mm)	0.050	0.10	0.050	0.15	0.20	0.25	0.25

The decisive distance computed with a unit floor, in the unit of the characteristic.

Table S19 A new family: distances in the floor of the produced family, and a baseline from the drawing nominal

Rule	target family's floor	source family's floor
M0	108	104
M1	35	33
M1s	122	121
Mc	38	43
M2	51/16	41/26
M3	49/21	40/17
M5	38/28	35/31
NN	131	130
<i>Wall angle only: decisive/safe distance; median and 90 % absolute error</i>		
nominal + offset	2.0	0.3; 1.8
M0	28	14.0; 27.1
M1	16	5.3; 15.9
M1s	123	121.1; 123.0
Mc	19	15.3; 18.3
M2	10.0/0.05	1.0; 4.5
M5	7.3/3.2	3.6; 4.6
M1n	129	126.0; 129.0
NN	130	125.9; 129.8

Upper block: all characteristics, δ_d/δ_s (floors). Lower block: wall angle of a new family (18 cases); nominal + offset: the design angle of the new family plus the median deviation q_{95} – design angle of the produced family.

Table S20 Calibration budget: distances and wrong verdicts at 10 floors by the relation of the new design to the produced alternatives

Relation	Rule	$k = 1$		$k = 2$		$k = 4$		$k = 6$	
		δ_d/δ_s	wrong	δ_d/δ_s	wrong	δ_d/δ_s	wrong	δ_d/δ_s	wrong
sibling	M1	3.5	0	12	7	17	14	18	15
sibling	M2	3.5/3.5	0	15/2.2	0	23/1.0	0	23/0.50	0
sibling	M5	–	–	14/1.6	0	8.1/1.9	0	7.6/1.2	0
sibling	NN	3.5	0	3.5	0	3.4	0	3.4	0
interpolation	M1	–	–	11	8	13	15	11	9
interpolation	M2	–	–	15/4.8	0	25/3.8	0	18/2.7	0
interpolation	M5	–	–	19/3.7	0	22/2.9	0	19/2.8	0
interpolation	NN	–	–	5.0	0	5.3	0	4.7	0
extrapolation	M1	29	26	25	28	25	28	25	28
extrapolation	M2	15/15	13	15/11	8	20/9.3	4	18/9.9	4
extrapolation	M5	–	–	18/11	6	19/8.8	3	19/8.9	3
extrapolation	NN	11	8	10	6	9.5	4	9.5	4

All subsets of k produced alternatives of the family. Sibling: the subset holds an alternative at the new design's force; interpolation: no sibling, forces on both sides; extrapolation: no sibling, the new force outside. Wrong: share of wrong verdicts at $|d| = 10$ (%). M5 from $k = 2$.

Table S21 Where the distances come from: decisive/safe distances (floors) of subsets and variants of the cases

Cases	NN	M1s	M2	M5	M5n
<i>New variant, siblings resolvably different</i>					
neither (58)	0.85	6.9	22/0	6.1/0	5.2/0
one (34)	2.5	5.0	25/0.30	6.1/0	7.2/0.20
both (34)	4.9	4.6	15/1.4	9.7/2.3	8.7/2.9
<i>New process setting, held-out force</i>					
100 kN, slower stroke (42)	11	15	17/8.1	18/8.1	16/9.7
300 kN, interpolated (42)	4.7	19	18/2.7	19/2.8	16/0
500 kN (42)	4.9	19	19/10.0	20/9.7	18/2.4
<i>Other family's alternatives added to the calibration</i>					
new variant (126)	3.4	15	53/0	11/0.05	9.8/0.20
new process setting (126)	8.5	20	53/1.0	28/5.4	16/2.7

Siblings resolvably different: effect-to-scatter ratio of at least one between the held-out alternative and its sibling. Pooled: the alternatives of the other geometry are added to the calibration set (Section 4.4).

S4 Learners

Table S22 repeats the part-level rules under other learners and settings, among them a setting selected by its leave-one-out error, and Table S23 shows how often a hyperparameter of the Gaussian processes sits at a bound.

Table S22 Part-level rules M3 and M3n under other learners and settings (decisive/safe, floors)

Learner	New variant		New setting		New family	
	M3	M3n	M3	M3n	M3	M3n
reference	23/0	11/0.05	31/16	13/7.4	49/21	133/128
regularised	23/0.30	13/0.05	27/14	13/7.7	49/26	133/128
flexible	27/0.05	11/0.10	36/13	14/7.6	49/26	133/129
ridge	32/0.35	13/0	37/7.8	16/5.0	51/21	133/128
quantile loss	26/0	12/0.15	30/15	15/8.5	50/27	133/129
selected	20/0.05	12/0.05	28/17	13/7.8	49/26	133/129

Reference: histogram gradient boosting, depth 3, 200 iterations, rate 0.08. Regularised: depth 2, 100 iterations, rate 0.05, L2 penalty 1. Flexible: depth 6, 400 iterations, rate 0.1. Ridge: linear regression on standardised inputs. Quantile loss: boosting of the 95th percentile. Selected: the boosting setting with the smallest leave-one-out error. Intervals: jackknife+.

Table S23 Gaussian-process fits with a hyperparameter at a bound

Scope	M5			M5n		
	fits	noise (%)	length (%)	fits	noise (%)	length (%)
New variant	126	97	52	126	99	32
New setting	42	100	69	42	100	83
Pooled variant	126	84	56	126	87	64
New family	14	100	50	14	100	29

Share of the fits whose noise level sits at its lower bound, the variance between the series of one geometry and force, and with a length scale at a bound (0.1 or 100).

S5 Procedure, guards and requirement scenarios

Table S24 gives the guard bands of step 6 and the operating characteristics of the procedure’s decision rule, Table S25 the performance of the applicability guards, Table S26 the verdicts at requirements anchored in capability, Table S27 those at requirements from general tolerances and Table S28 the cheapest rule in the tolerance scenario as a function of the costs.

Table S24 The procedure’s decision rule: guard band and operating characteristics

Relation	Rule	g	rule alone	step 6	trial at 10 floors (%)
variant	M1s	2.0	4.9/4.9	6.9/2.9	0 → 1
	M2	0	23/0.15	23/0.15	33 → 33
	M5	0	7.2/0.75	7.2/0.75	1 → 1
	M5n	0	7.2/0.55	7.2/0.55	0 → 0
	NN	0.25	3.4/3.4	3.6/3.1	0 → 0
	M2n	0	19/0	19/0	23 → 23
interpolated	M3n	0	11/0.05	11/0.05	9 → 9
	M1s	∞	19/19	trial	0 → 100
	M2	0.75	18/2.7	19/2.0	25 → 31
	M5	0.50	19/2.8	20/2.3	39 → 40
	M5n	0	16/0	16/0	25 → 25
	NN	2.0	4.7/4.7	6.7/2.7	0 → 0
extrapolated	M2n	0	13/0	13/0	19 → 19
	M3n	6.0	15/7.0	21/1.0	24 → 55
	M1s	∞	19/19	trial	0 → 100
	M2	∞	18/9.9	trial	25 → 100
	M5	∞	19/8.9	trial	26 → 100
	M5n	6.8	18/8.6	25/1.9	19 → 48
new family	NN	7.0	9.5/9.5	17/2.5	0 → 26
	M2n	7.3	16/9.1	23/1.9	18 → 50
	M3n	6.5	13/7.8	19/1.3	19 → 47
	M1s	∞	122/122	trial	0 → 100
	M2	∞	51/16	trial	26 → 100
	M5	∞	38/28	trial	4 → 100
	M5n	∞	132/130	trial	6 → 100
	NN	∞	131/131	trial	0 → 100
	M2n	∞	132/129	trial	17 → 100
	M3n	∞	133/128	trial	10 → 100

g : smallest observable margin (floors) beyond which at least 95 % of the decided verdicts are correct for requirements on the grid within 20 floors of the truth; ∞ : never, the procedure sends every design to trial. Distances δ_d/δ_s in floors.

Table S25 Applicability guards: distances and wrong verdicts among the cases a guard lets through

Scope	Guard	Guarded	M2		M5		NN	
			δ_d/δ_s	wrong	δ_d/δ_s	wrong	δ_d	wrong
New setting	none	0	18/8.1	6/239	19/6.8	4/239	8.5	6/239
	range	84	$\infty/0$	0/80	$\infty/0$	0/80	∞	0/80
	novelty	84	$\infty/0$	2/80	$\infty/0$	2/80	∞	0/80
New family	none	0	51/16	23/239	38/28	44/239	131	106/239
	range	126	$\infty/0$	0/0	$\infty/0$	0/0	∞	0/0
	novelty	114	$\infty/0$	0/24	$\infty/0$	6/24	∞	6/24

Guarded: cases (of 126) sent to trial. Range guard: geometry not calibrated or force outside the calibrated range; novelty guard: simulated q_{95} outside the calibration range. Wrong: wrong verdicts at $|d| = 10$ among those let through.

Table S26 Requirements anchored in capability: right and trial verdicts (%)

Relation	Rule	$P_{pk} = 1$	$P_{pk} = 1.33$	$P_{pk} = 1.67$	$P_{pk} = 2$
New variant	NN	82/0	91/0	93/0	96/0
	M1s	74/0	81/0	84/0	90/0
	M1	60/0	67/0	68/0	73/0
	M2	17/82	20/79	27/73	33/67
	M5	26/69	44/53	56/43	67/33
	M5n	27/68	40/56	55/44	67/33
New setting	NN	60/0	65/0	75/0	79/0
	M1s	73/0	79/0	83/0	85/0
	M1	51/0	52/0	58/0	63/0
	M2	29/55	32/53	42/44	47/43
	M5	29/53	32/52	37/48	39/50
	M5n	27/60	33/63	38/59	43/55
New family	NN	50/0	51/0	53/0	53/0
	M1s	43/0	45/0	48/0	48/0
	M1	54/0	54/0	57/0	60/0
	M2	29/52	33/51	37/48	39/45
	M5	48/17	49/21	52/20	55/18
	M5n	50/2	50/2	50/3	52/2

Upper limit at centre + $3P_{pk}$ standard deviations of each alternative's parts; every alternative meets it, so the rest of the verdicts are false rejects. The limits lie a median 1.0, 1.7, 2.5, 3.1 floors above q_{95} .

Table S27 Verdicts against requirements from general angular tolerances (ISO 2768-1)

Rule	New variant (%)				New setting (%)				New family (%)			
	right	<5	FA	trial	right	<5	FA	trial	right	<5	FA	trial
M0	61	69	39	0	61	69	39	0	61	69	39	0
M1	83	62	9	0	75	53	13	0	61	62	30	0
M1s	100	100	0	0	89	72	9	0	31	44	65	0
M2	67	25	0	33	72	38	4	25	44	47	0	39
M5	93	75	0	7	72	38	4	25	62	44	9	16
NN	100	100	0	0	97	91	0	0	31	44	65	0
M5n	94	78	0	6	86	59	0	14	31	44	62	2

Wall and arm angle, classes f/m, c and v: 108 decisions per rule and scope, 69 with a failing alternative, 32 with the limit within 5 floors of the truth. right: correct verdicts; <5: correct among the decisions within 5 floors; FA: false accepts among the failing alternatives; trial: sent to trial.

Table S28 Cheapest rule in the tolerance scenario by the costs of a false reject and a trial

Relation	c_{FR}	0.02	0.05	0.1	0.2	0.3	0.5
New variant	0.25	M1s/NN	M1s/NN	M1s/NN	M1s/NN	M1s/NN	M1s/NN
	0.5	M1s/NN	M1s/NN	M1s/NN	M1s/NN	M1s/NN	M1s/NN
	1	M1s/NN	M1s/NN	M1s/NN	M1s/NN	M1s/NN	M1s/NN
	2	M1s/NN	M1s/NN	M1s/NN	M1s/NN	M1s/NN	M1s/NN
New setting	0.25	M2n/M3n	M2n/M3n	NN	NN	NN	NN
	0.5	M2n/M3n	M2n/M3n	M2n/M3n	NN	NN	NN
	1	M2n/M3n	M2n/M3n	M2n/M3n	M2n/M3n	M2n/M3n	NN
	2	M2n/M3n	M2n/M3n	M2n/M3n	M2n/M3n	M2n/M3n	M2n/M3n
New family	0.25	trial	trial	M2	M2	M5	M5
	0.5	trial	trial	trial	M2	M5	M5
	1	trial	trial	trial	trial	M5	M5
	2	trial	trial	trial	trial	trial	M5

Costs relative to a false accept; columns: cost of a trial c_T . Ties joined by a slash. trial: a physical trial for every design, the current-practice baseline.

S6 Robustness

Table S29 repeats the evaluation under variants of the data, the constants and the model specifications, and gives the intervals of the refitting bootstrap. Table S30 tests whether the process signals recorded for every part explain the drift that sets the floor. The sensitivities of the simulation over the geometric corners of the simulation dataset, which have no production counterpart, and the part-level in-line verifiability of the characteristics from the press-force record are provided with the code but not used in the paper.

Table S29 Decisive/safe distances (floors) under the robustness variants

Variant	New variant				New setting			New family	
	M1s	M2	M5	NN	M2	M5	NN	M2	M5
Reference	4.9	23/0.15	7.2/0.75	3.4	18/8.1	19/6.8	8.5	51/16	38/28
Temperature-adjusted	5.0	23/0.10	7.2/0.65	3.4	17/7.6	19/6.8	9.4	48/15	37/27
First 150 parts dropped	6.5	27/0.20	9.0/0.45	4.1	21/8.7	24/8.7	12	63/18	39/29
Smoothed simulation	4.8	25/0.30	7.2/0.70	3.4	18/7.6	20/6.9	8.5	49/25	37/27
Adequacy share 0.90	4.9	23/0.20	7.2/0.70	3.3	18/7.5	18/6.4	8.4	51/15	38/28
Adequacy share 0.99	5.1	23/0	7.5/0.70	3.3	18/8.3	20/6.9	8.7	51/16	38/28
Floor: batches of 25	4.6	21/0.15	6.4/0.70	2.9	16/6.9	17/6.3	7.7	43/14	31/23
Floor: batches of 100	5.3	24/0.15	8.2/0.80	3.5	19/8.5	21/7.9	8.9	57/17	46/29
Floor: 90 % quantile	6.2	31/0.20	8.8/0.90	4.0	21/10	23/8.6	10	61/20	45/34
Floor: 99 % quantile	3.4	14/0.10	5.2/0.50	2.4	11/4.4	13/4.4	5.7	37/11	29/15
Floor between series	2.9	11/0	1.3/0.20	0.75	9.9/2.6	11/2.3	2.7	16/3.8	12/9.3
Target 90 %	4.3	17/0	5.8/0	1.9	16/5.2	17/4.0	5.4	45/9.6	23/16
Target 99 %	7.8	28/1.4	11/3.6	5.1	27/11	34/11	11	60/28	60/34
Calibration series of 50	5.0	22/0.20	6.8/0.70	3.4	18/7.5	19/6.2	8.4	50/16	37/28
Calibration series of 100	4.9	22/0.40	6.8/0.40	3.3	18/7.5	19/6.2	8.7	50/16	38/28
Calibration series of 250	4.9	22/0.45	7.0/0.45	3.3	18/7.5	19/6.2	8.7	50/16	38/28
Interval level 0.80		23/0.15	6.4/1.0		18/8.1	17/8.1		50/17	38/28
Interval level 0.95		23/0.15	8.0/0.25		18/8.1	20/6.2		51/16	39/27
GP: Matérn 5/2			7.2/0.75			19/6.8			38/28
GP: linear trend			7.2/0.70			19/6.0			38/28
GP: one-hot lubrication			7.4/0.95			19/6.9			38/28
GP: length-scale floor			7.3/0.60			19/6.2			38/28
GP: no noise floor			6.7/1.5			18/7.4			38/30
GP: sampling noise floor			6.7/1.4			18/7.4			38/30
Unit: q_{95} reproducibility	5.3	17/0.15	6.9/0.40	2.8	15/6.0	18/6.2	6.9	41/13	34/30
Leave one pattern out	4.9	22/0.30	7.0/0.80	3.4					
Refit bootstrap, δ_d	4.9–7.3	22–23	4.0–11	2.3–4.6	16–18	18–20	6.9–9.2	49–53	34–39

Alternative-level rules refitted for every variant; empty cells: the variant does not affect the rule. Floor variants rescale the distances with other floors. Calibration series of n : the calibration alternatives' q_{95} from their first n parts. Leave one pattern out: columns M1s, M2, M5 and NN of the first block. Unit: distances in units of the 95 % quantile of the difference between the q_{95} of two batches of 100 parts instead of F . Refit bootstrap: 95 % interval when the lubrication patterns of each geometry are resampled and every rule is refitted (200 resamples).

Table S30 Drift of the batch centres explained by the process signals

Characteristic	Batches	R^2 , out of fold	F	F , drift removed	Change (%)
Draw-in, mid-side (mm)	177	0.20	0.051	0.046	−9
Draw-in, corner (mm)	177	0.07	0.047	0.045	−4
Flange waviness (mm)	177	−0.15	0.0028	0.0030	+6
Wall angle (°)	177	−0.12	0.076	0.086	+12
Arm angle, cut (°)	177	−0.23	0.11	0.12	+11
Cup depth (mm)	177	0.11	0.011	0.0092	−16
Bottom dome, cut (mm)	177	0.10	0.0070	0.0063	−10

Ridge regression of the batch centres on the batch centres of press force, punch temperature, sheet thickness and oil film, folds by alternative; F recomputed after removing the predicted drift.

References

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